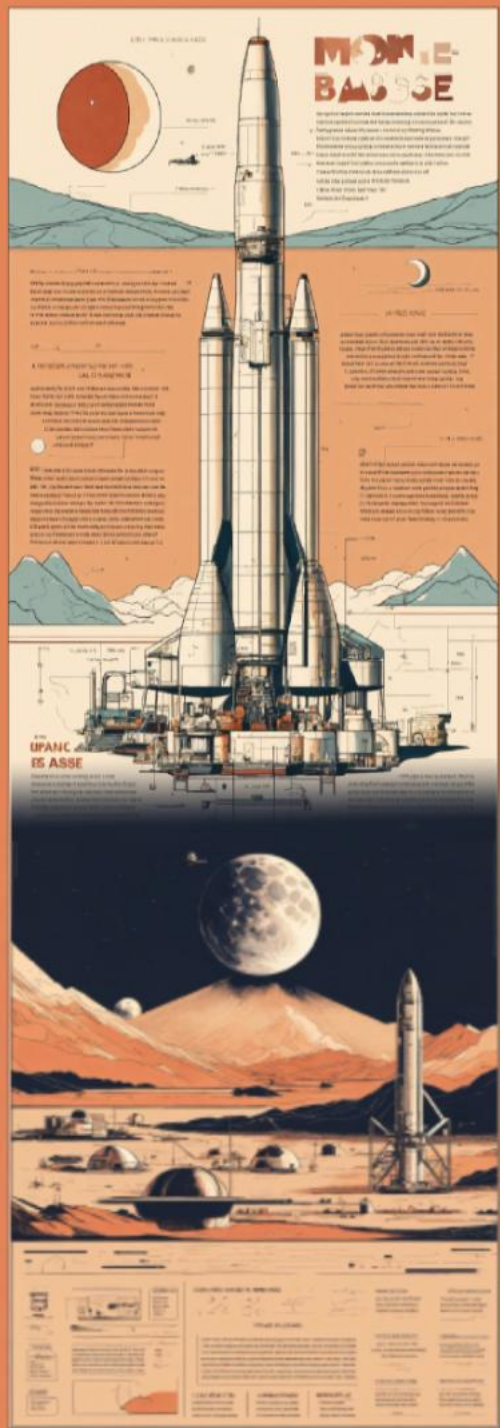




APRICOT

Apricot

Analysis Integration Demo

2026.09.25 - OpenMBEE Dev. Meeting

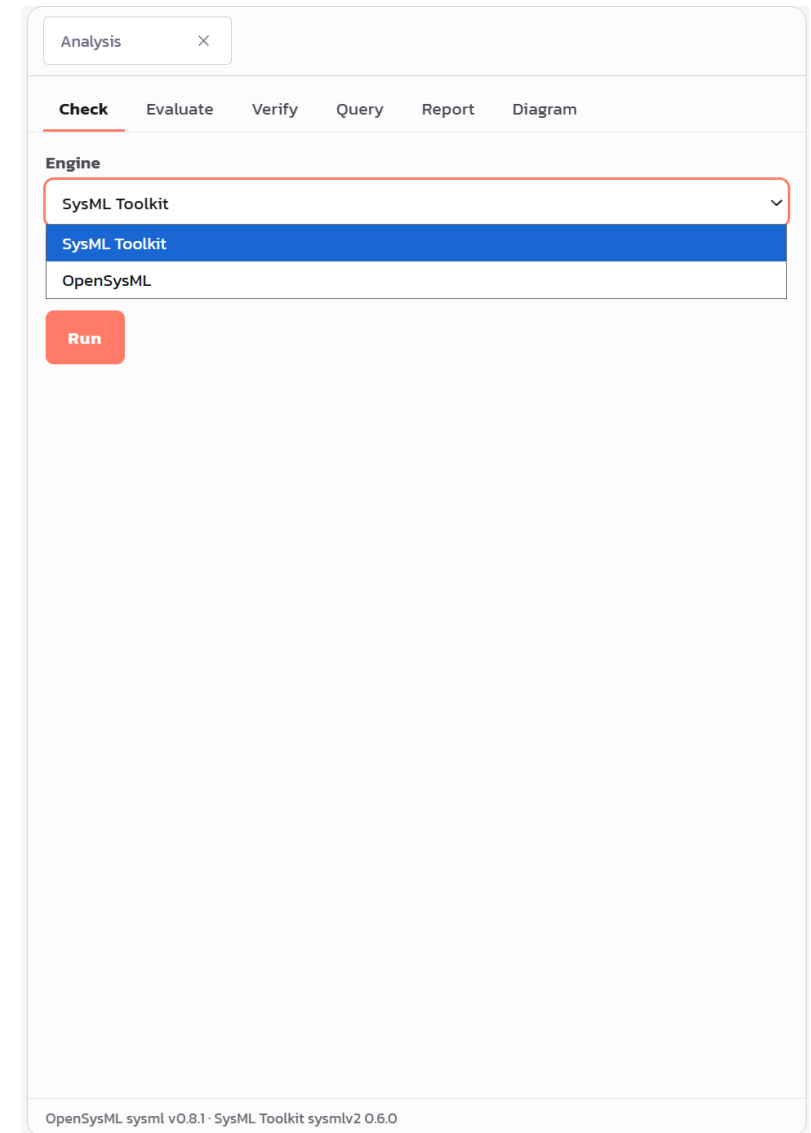
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METADEV

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4. Integration Architecture
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Intro

Apricot is a Web tool for SysML2

<https://apricot.tools>

We are integrating SysML2 engines in to it:

- **OpenSysml**
- **Sysml-Toolkit**

Work in progress...

Preliminary results:

<https://github.com/metadevpro/apricot-demo-integrations>



Runtimes

SysML Toolkit

Repository:

<https://github.com/Open-MBEE/sysml-toolkit>

Created by Blake Regalia, Robert Karban, Johannes Gross. **Planetary Utilities**



Implemented in **Rust**.

License: Apache 2.0



OpenSysML



OpenSysML

Repository:

<https://github.com/Open-MBEE/OpenSysML>

<https://opensysml.org/>

Created by Jason Han, Luis Carlos, et al.
NASA/JPL



Implemented in **Go**.

License: Apache 2.0



Goals

1. Tests both runtimes
2. Early demo on what we are building
3. Easy way to compare in different runtimes
4. Have a set of examples useful to explain
 - Strong features of each runtime
 - Weak features or errors
 - Out of scope for a runtime
5. Integrate Apricot with open-source tools

Demo. Delivery Quadcopter

Built a simple demo domain for a Quadcopter.

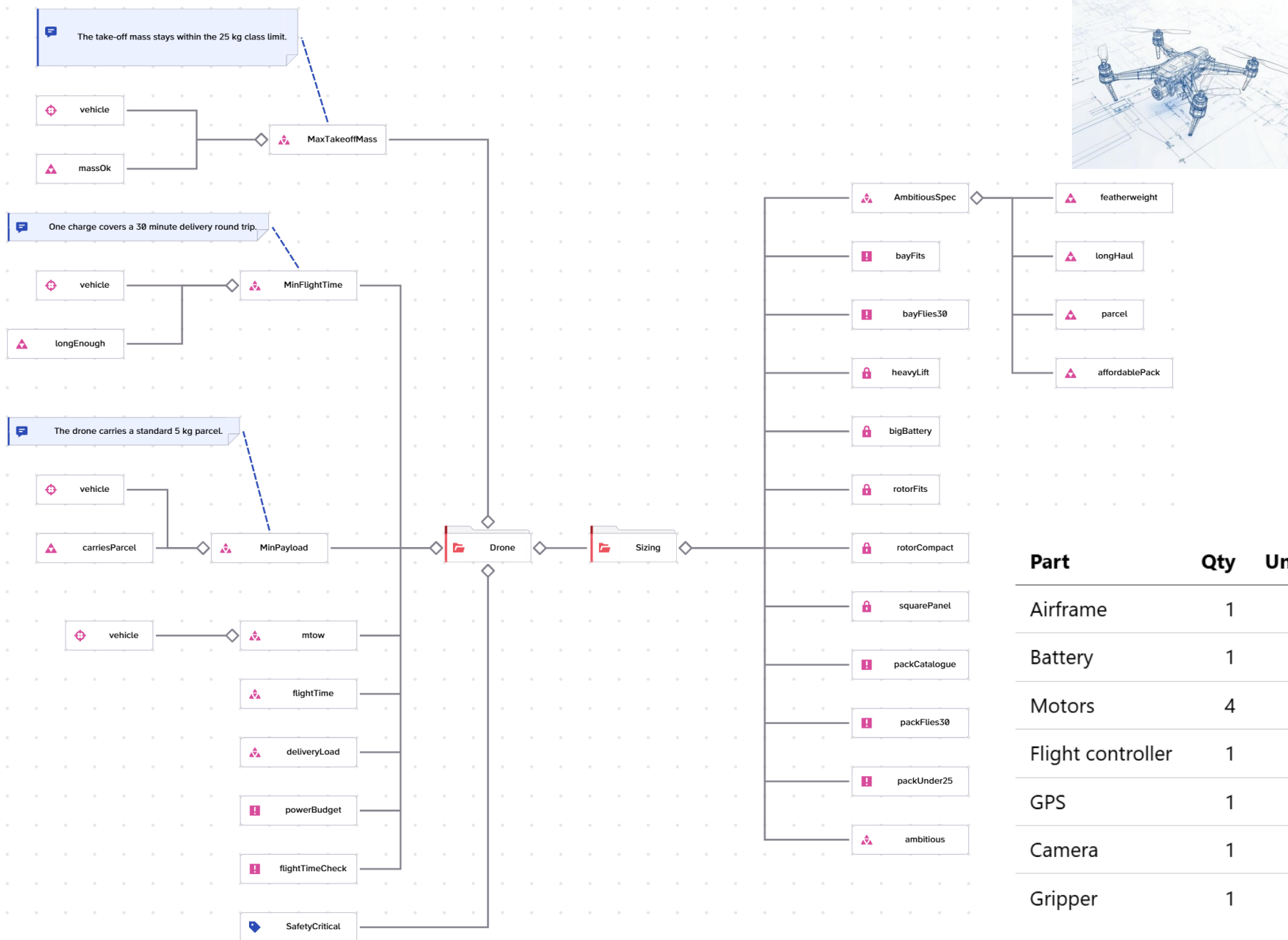
File: **drone.sysml12**

Described in terms of:

- Components
- Requirements
- Costs
- Autonomy
- Weight
- Constrains & tradeoffs

Illustrative enough to showcase the runtimes.





Part	Qty	Unit mass	Power draw	Cost
Airframe	1	3.2 kg	—	450 €
Battery	1	4.4 kg	1100 Wh	900 €
Motors	4	0.55 kg	—	160 €
Flight controller	1	0.25 kg	12 W	380 €
GPS	1	0.1 kg	2 W	120 €
Camera	1	0.6 kg	15 W	700 €
Gripper	1	0.4 kg	5 W	150 €

Check

SysML Toolkit

Lenient: sum used without its import is only a warning (it fails only with --strict).

Checks syntax and name resolution.

Demo model: valid, no findings.

OpenSysML

Strict: the same sum is an error

(“did you mean RealFunctions::sum?”), so a model the Toolkit accepts can be rejected here.

Same check, with its own parser.

Demo model: valid, no errors.

Evaluate

SysML Toolkit

No instantiation: `motors[4]` stays one element, so `sum(motors.mass) = 0.55 kg`, not 2.2

The redefinition part `:>> motors {...}` drops the inherited `[4]`: `size(motors) = 1`

Motor **0.55 kg** - Takeoff 14.5 kg - Endurance 35.6 min

The error propagates to every value that depends on mass.

```
Drone::drone::motorMass
Drone::drone::takeoffMass
Drone::drone::hoverPower
Drone::drone::endurance
```

OpenSysML

Materializes the 4 motors:

`motors.mass = [0.55, 0.55, 0.55, 0.55]`, `sum = 2.2 kg` (correct)

Motor **2.2 kg** - Takeoff 16.15 kg - Endurance 32.0 min

Executes the model: builds the drone with all its parts

```
Drone::drone.motorMass
Drone::drone.takeoffMass
Drone::drone.hoverPower
Drone::drone.endurance
```

Verify

SysML Toolkit

False pass: with a 6.5 kg parcel, endurance 32.3 min, so R-02 is satisfied (it inherits the 1-motor error)

Checks all constraints automatically; missing data gives “undecided”; no confidence reported
Z3 substitutes derived attributes: proves the 5000 Wh battery impossible and returns $\sqrt{2}$ exactly

```
powerBudget (AssertConstraintUsage): satisfied  
flightTimeCheck (AssertConstraintUsage): satisfied  
reserve (AssertConstraintUsage): satisfied  
massOk (...satisfies R-01): satisfied  
longEnough (...satisfies R-02): satisfied  
carriesParcel (...satisfies R-03): satisfied  
6 satisfied, 0 violated, 23 undecided
```

OpenSysML

Z3 %check treats the derived takeoffMass as a free unknown: false SAT for a 5000 Wh battery
Refuses $\sqrt{2}$: its evaluator has no rational for the solver’s value

6.5 kg parcel: endurance 29.4 min, R-02 violated (correct)

In the panel, checks only the constraints you name (-satisfy checks them all, API only); reports confidence per verdict

```
Drone::powerBudget  
Drone::flightTimeCheck  
Drone::Sizing::heavyLift
```

Query

SysML Toolkit — KerML

Total cost query returns 2860 € instead of 3340 €: the same 1-motor error inside the query

Expressions that compute: counts, sums, margins (R-02 endurance margin = 5.58 min)

Lists the satisfy usages only as anonymous elements: cannot answer “who satisfies R-01?”

```
size(ownedMember(Drone))  
  
ownedMember(Drone::Sizing)  
  
Drone::drone.endurance  
  
Drone::drone.battery istype Drone::Motor
```

OpenSysML — OSLC

Cannot read attribute values: sysml:mass is rejected, so it sees motors = 4 but cannot compute with it

No property for satisfy relationships: cannot answer “who satisfies R-01?”

Filters by type, name and owner (lists R-01..R-03).

Reports malformed queries instead of returning nothing.

```
oslc.where=rdf:type="RequirementUsage"&  
oslc.select=sysml:shortName,sysml:type  
  
oslc.where=sysml:type="Drone::Motor"&  
oslc.select=sysml:multiplicityLower,sysml:multiplicityUpper  
  
oslc.where=rdf:type="PartUsage" and  
sysml:owner="Drone::drone"&oslc.select=sysml:name&  
oslc.orderBy=-sysml:name
```

Report

SysML Toolkit

Not available.

Feature specific for OpenSysML.

OpenSysML

Relies on its own DocumentQueries library, not part of the **OMG standard library**: the report does not port to other engines

Renders Markdown or HTML: bill of materials, critical parts, requirements and who satisfies them.

Computed from the model: change a mass and the table changes.

```
part def DesignReport :> Document {  
  ...  
}
```

Diagram

SysML Toolkit

Draws motors : **Motor [4]** and every port, yet computes with one motor.

PlantUML views: interconnection (7 parts, ports, 6 links), state, action, tree.

The Diagram panel always calls the Toolkit.

```
@startuml
...
@enduml
```

OpenSysML

Its interconnection drops the **[4]** on motors and links parts without ports, yet computes with four motors.

Renders text, mermaid, markdown, DOT or **PlantUML**, only through the API (-render), not in the Diagram panel.

Not integrated in Apricot yet.

Strengths of each engine

SysML Toolkit v. 0.6.0

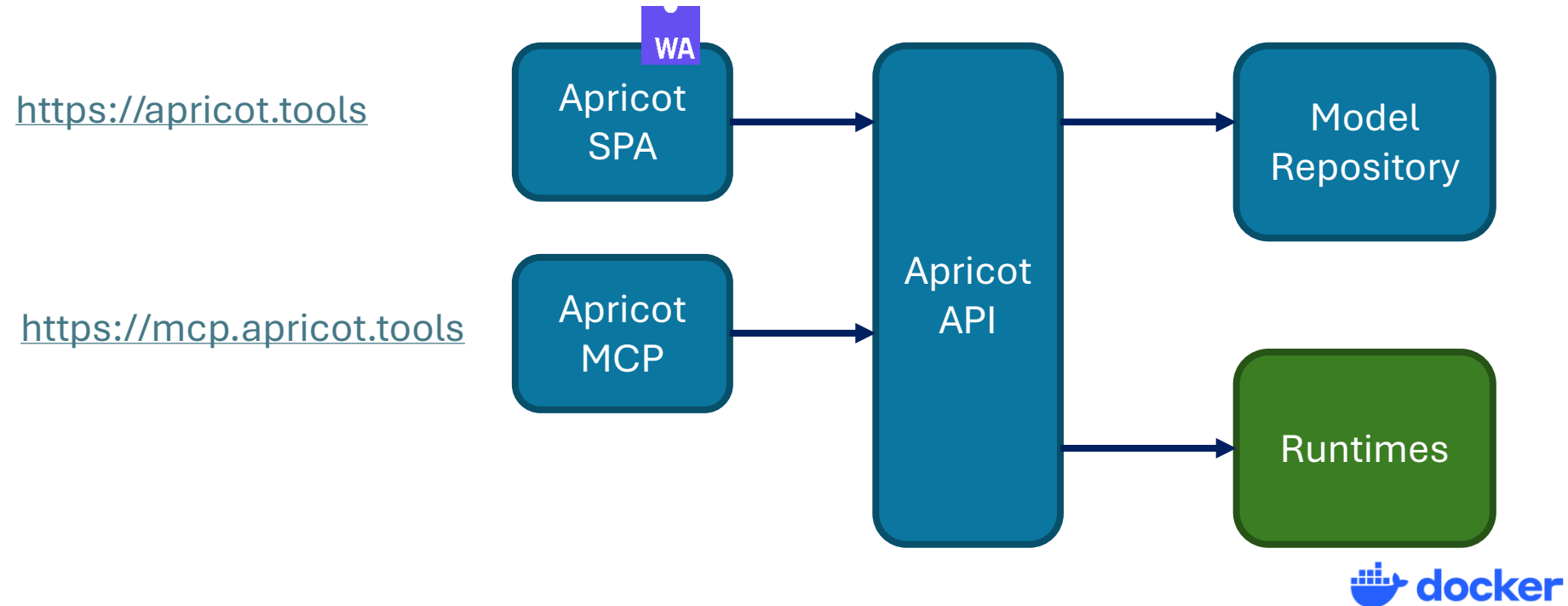
- Standard interchange as files: KerML 10.4 JSON (compact/full), CBOR, .kpar, normative element ids
- Z3 reasons over the formulas: substitutes derived attributes
- Queries are KerML expressions that compute values
- Narrows open values to ranges without a solver
- Formatter, linter, refactorings, LSP
- WASM for the browser

OpenSysML v. 0.8.1

- Executes the model: instantiates parts as their multiplicity declares
- Z3 explains and optimizes (looks for optimization)
- Implements reporting
- Provides confidence per verdict
- REPL
- SysML v2 API sync
- LSP
- Client libraries for Go, Python, Node/T

Parsing is now on par: both parse the same 391/396 files of the 2026-03 SysML2 corpus.

4. Integration Architecture



5. Conclusions

- Work in progress...
- Available at: <https://github.com/metadevpro/apricot-demo-integrations>
- Quick integration with 3rd party opensource components
- Faster than expected (less than one week)
- The integration provides a **quick way to test and compare both runtimes**
- Evaluating to incorporate more verification/calc tools to Apricot



Next-gen SysML2 tooling



Know more

Contact

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Open Beta

Try it at

<https://apricot.tools>